

Design for class “IPaymentQRCode”

Description:

IPaymentQRCode defines the public contract of the QR-code payment subsystem. It provides generalized behaviors for QR-based payment gateways, independent from any specific provider such as VietQR or VNPay. The interface isolates the rest of AIMS from implementation details inside the subsystem.

Table 1. Attribute design

No attributes.

Table 2. Operation design

	Name	Return type	Description
1	generateQRCode	QRCode	Generate a QR code for the given order so that the customer can scan and pay.
2	checkPaymentStatus	PaymentStatus	Check and return the current payment status of the given order.

1) generateQRCode(order: Order): QRCode

Parameter:

- order: Order — current order must containing payment amount, order ID, and payment content.

Exception:

- PaymentException if QR generation fails or gateway communication is unsuccessful.

Method:

- Extract required payment data from order.
- Build request data for VietQR.
- Request QR code from the gateway.
- Parse the response and return a QRCode object.

2) checkPaymentStatus(order: Order): PaymentStatus

Parameter:

- order: Order — current order whose payment status must be checked.

Exception:

- `PaymentException` if status retrieval fails or response is invalid.

Method:

- Extract order ID from order.
- Build status-check request.
- Send request to VietQR.
- Parse the response and return `PaymentStatus`.